

Lab 4: ICOW Model - Risk Analysis

CEVE 421/521

2026-02-06

1 Overview

On Wednesday, you met the **Island City On a Wedge (ICOW)** model and estimated city geometry parameters using Google Maps. Today you'll run the model to analyze flood risk for a city with a modest dike — enough protection to make things interesting, but not enough to stop big storms.

We'll use two complementary approaches:

1. **Stochastic simulation:** Sample specific storms and compute individual damage outcomes
2. **Expected Annual Damage (EAD):** Integrate over the surge distribution analytically

This is the same hazard-damage convolution from Lab 2, but applied to a realistic city-scale model instead of a single house.

Learning Objectives:

1. Understand the ICOW model and how it computes flood damage for a simplified coastal city
2. Apply hazard-damage convolution (from Lab 2) to a realistic city-scale model
3. Compare stochastic simulation (distribution of outcomes) vs. EAD (expected value via integration)

References:

- ICOW.jl documentation
- SimOptDecisions documentation

2 Before Lab

Complete these steps **BEFORE** coming to lab:

1. **Accept the GitHub Classroom assignment** (link on Canvas)
2. **Clone your repository:**

```
git clone https://github.com/CEVE-421-521/lab-04-S26-yourusername.git
cd lab-04-S26-yourusername
```

3. **Open the notebook** in VS Code or Quarto preview
4. **Run the first code cell** — it will automatically install any missing packages (this may take 10-15 minutes the first time)

! Important

Submission: You will render this notebook to PDF and submit the PDF to Canvas. See Section 10 for details.

3 Setup

3.1 Load Packages

First, we load custom packages that have been developed for this class. These are installable Julia packages, but are not (yet) on the “registry” of Julia packages, so they need to be installed directly from GitHub. This motivates some slightly more complicated package-loading code.

```
# install SimOptDecisions and ICOW
try
    using SimOptDecisions
    using ICOW
catch
    import Pkg
    try
        Pkg.rm("SimOptDecisions")
    catch
    end
    try
        Pkg.rm("ICOW")
    catch
    end
    Pkg.add(; url="https://github.com/dossgollin-lab/SimOptDecisions")
    Pkg.add(; url="https://github.com/dossgollin-lab/ICOW.jl")
    Pkg.instantiate()
    using SimOptDecisions
    using ICOW
end
using ICOW.Stochastic
using ICOW.EAD
```

```
using CairoMakie
using Distributions
using Random
using Statistics
using DataFrames
```

```
Random.seed!(2026)
```

```
TaskLocalRNG()
```

ICOW has two submodules: `ICOW.Stochastic` for event-by-event simulation (with random dike failure), and `ICOW.EAD` for analytical expected damage via integration.

4 Meet the Model

Before running any code, let's build a mental model of what ICOW computes.

4.1 The City

The default configuration represents Manhattan — the same parameters from Wednesday's lecture.

```
config = StochasticConfig()
```

```
StochasticConfig{Float64}
  Geometry      V_city=1.5e12 H_bldg=30.0 H_city=17.0 D_city=2000.0
W_city=43000.0 H_seawall=1.75
  Dike          D_startup=2.0 w_d=3.0 s_dike=0.5 c_d=1000.0
  Zones         r_prot=1.1 r_unprot=0.95
  Withdrawal    f_w=1.0 f_l=0.01
  Resistance    f_adj=1.25 f_lin=0.35 f_exp=0.115 t_exp=0.4 b_basement=3.0
  Damage        f_damage=0.39 f_intact=0.03 f_failed=1.5 t_fail=0.95 p_min=0.05
f_runup=1.1
  Threshold     d_thresh=4.0e9 f_thresh=1.0 gamma_thresh=1.01
```

Recall the key geometry parameters:

Parameter	Default (Manhattan)	Meaning
V_{city}	\$1.5 trillion	Total city value
H_{city}	17 m	Peak elevation above sea level
H_{seawall}	1.75 m	Existing seawall height

4.2 The Baseline Policy

On Wednesday, you learned about ICOW's five defense levers: Withdrawal (W), Dike Base (B), Dike Height (D), Resistance Height (R), and Resistance Fraction (P).

For this lab, we'll give the city a modest dike ($a_{\text{frac}}=0.15$ allocates 15% of the city's elevation as dike height, or about 2.5m). This is enough to stop smaller storms but can fail during larger ones — which makes the stochastic simulation interesting.

```
baseline = StaticPolicy(;
  a_frac=0.3, # use 30% of city elevation as protection budget
  w_frac=0.1, # 10% of budget goes to withdrawal
  b_frac=0.2, # 20% of remaining budget goes to dike base
  r_frac=0.1, # resistance height = 10% of H_city
  P=0.5,      # flood-proof 50% of buildings
)
fd = FloodDefenses(baseline, config)
fd
```

5 Stochastic Analysis: One Storm

Let's start with the simplest case — one specific surge height. Even a single surge produces a *distribution* of damages because of stochastic dike failure.

5.1 Run One Storm

```
scenario = StochasticScenario(  
  surges=[4.0],  
  mean_sea_level=[0.0],  
  discount_rate=0.0  
)  
outcome, trace = SimOptDecisions.simulate_traced(  
  config, scenario, baseline  
)
```

Line 6

`simulate_traced` returns both the outcome and a trace object that records what happened at each time step.

5.2 Inspect the Trace

The trace records what ICOW computed internally. Let's look at the step record for our single storm:

```
record = trace.step_records[1]
```

```
(investment = 5.564637817575113e10, damage = 1.955224393832893e9, W = 0.51, R =  
1.7000000000000002, P = 0.5, D = 3.6719999999999997, B = 0.918)
```

Each record contains: investment (cost of building defenses), damage (flood damage), and the defense levels (W, R, P, D, B). Notice that D is nonzero — that's our modest dike.

5.3 Repeat Many Times

The damage from a 4-meter surge isn't deterministic — ICOW samples whether the dike fails using a random Bernoulli draw. Let's run the same 4m surge many times to see the distribution of outcomes:

```
damages_one_surge = map(1:1000) do _  
  outcome = simulate(config, scenario, baseline)  
  SimOptDecisions.value(outcome.damage)  
end
```

Line 1

`map` with a `do` block applies the function to each element of `1:1000`, returning a vector of results. Each call advances the global RNG (seeded above), so the dike failure draw is different each time.

Line 3

`SimOptDecisions.value()` extracts the numeric value from an outcome field.

5.4 Visualize

```
let
  fig = Figure(size=(600, 350))
  ax = Axis(
    fig[1, 1];
    xlabel="Damage (\$B)",
    ylabel="Count",
    title="Damage from a 4m surge (1,000 trials)"
  )
  hist!(ax, damages_one_surge ./ 1e9; bins=30)
  fig
end
```

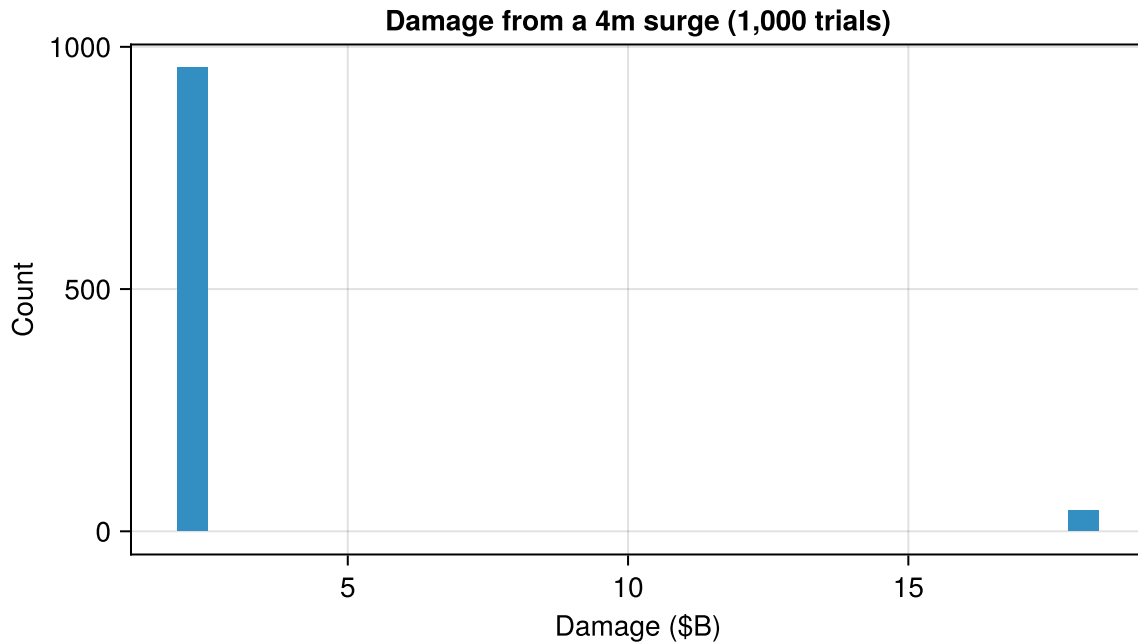


Figure 1: Distribution of damages from a 4-meter storm surge over 1,000 simulations. The two modes correspond to dike-intact and dike-failed outcomes.

💡 Question 1

The histogram in Figure 1 shows two clusters of damage values for a single surge height. Why does the same surge produce different damage amounts? What physical mechanism in the ICOW model creates this pattern?

6 Stochastic Simulation: Many Storms

Now let's sample many different surge heights from a GEV distribution — this is the full Monte Carlo convolution from Lab 2, applied to the ICOW model.

6.1 Sample Surges

```
surge_dist = GeneralizedExtremeValue(3.0, 1.0, 0.15)
n_samples = 500
surges = rand(surge_dist, n_samples)
```

This uses the same type of GEV distribution you worked with in Labs 2 and 3: location $\mu = 3.0$ m, scale $\sigma = 1.0$ m, and shape $\xi = 0.15$ (a heavy right tail).

6.2 Run Simulations

```
damages = map(surges) do surge
    scenario = StochasticScenario(
        surges=[surge],
        mean_sea_level=[0.0],
        discount_rate=0.0
    )
    outcome = simulate(config, scenario, baseline)
    SimOptDecisions.value(outcome.damage)
end
```

6.3 Visualize: Surge vs Damage

```
let
    fig = Figure(size=(800, 350))

    ax1 = Axis(
        fig[1, 1];
        xlabel="Storm Surge (m)",
        ylabel="Damage (\$B)",
        title="Surge vs. Damage"
    )
    scatter!(ax1, surges, damages ./ 1e9; markersize=4, alpha=0.5)

    ax2 = Axis(
        fig[1, 2];
        xlabel="Damage (\$B)",
        ylabel="Count",
        title="Damage Distribution"
    )
    hist!(ax2, damages ./ 1e9; bins=30)

    fig
end
```

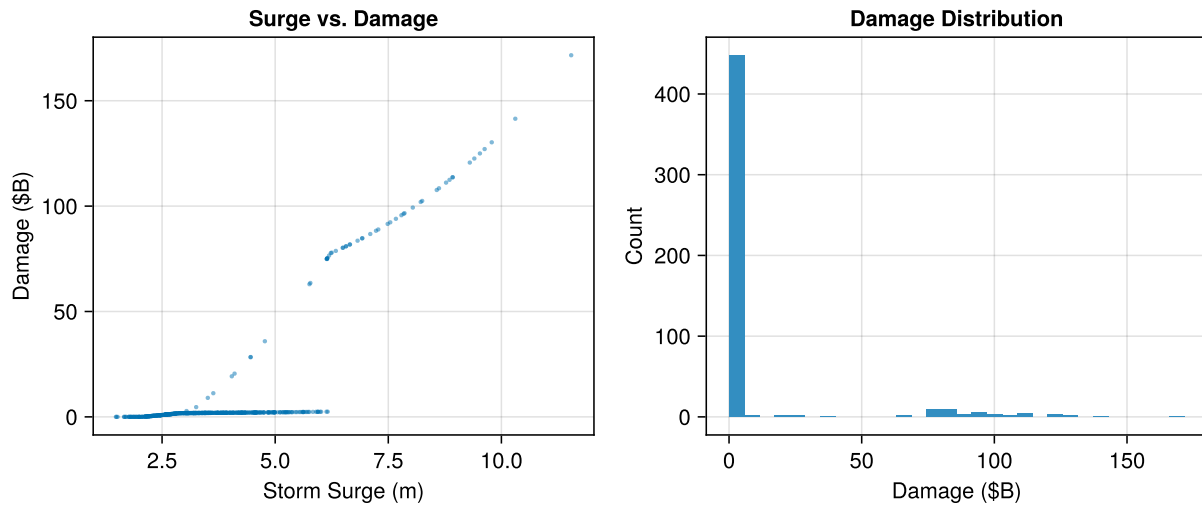


Figure 2: Stochastic simulation results. Left: two branches visible (dike intact vs. failed) merging at high surges. Right: most samples cluster at low damages — the tail is sparsely sampled.

💡 Question 2

Look at the scatter plot in Figure 2. You should see two “branches” — one where the dike holds and one where it fails. At what surge height do the branches merge (the dike always fails)? Also: notice how few samples appear in the high-damage tail of the histogram. What does this tell you about the efficiency of naive Monte Carlo for estimating tail risk?

6.4 Risk Statistics

```
DataFrame(
  Metric=[
    "Mean Annual Damage",
    "Prob(damage > \">$10B)",
    "Maximum Damage"
  ],
  Value=[
    "\ $$(round(mean(damages) / 1e9; digits=2))B",
    "$ (round(100 * mean(damages .> 10e9); digits=1))%",
    "\ $$(round(maximum(damages) / 1e9; digits=1))B"
  ]
)
```

Table 1: Summary risk statistics from stochastic simulation

Row	Metric	Value
	String	String
1	Mean Annual Damage	\$10.25B
2	Prob(damage > \$10B)	10.2%
3	Maximum Damage	\$171.6B

💡 Question 3

A city planner says: “Our expected annual damage is \$X billion, so we should budget \$X billion per year for flood recovery.” What’s wrong with this reasoning? Use the damage distribution in Figure 2 to support your answer.

7 EAD Mode: Integration Instead of Sampling

In the stochastic section, we sampled storms and averaged damages. EAD mode computes the same expected value by **integrating** over the surge distribution — no sampling noise.

This is the analytical counterpart to the stochastic mean: same GEV distribution, same damage function, but computed deterministically via numerical quadrature.

7.1 Run EAD

```
ead_config = EADConfig(
    n_years=1,
    integrator=QuadratureIntegrator(rtol=1e-6)
)
ead_scenario = EADScenario(
    surge_loc=3.0,
    surge_scale=1.0,
    surge_shape=0.15,
    mean_sea_level=[0.0],
    discount_rate=0.0
)
outcome_ead = simulate(ead_config, ead_scenario, baseline)
ead_quadrature = SimOptDecisions.value(outcome_ead.expected_damage)
```

7.2 Compare Stochastic vs. EAD

```
mc_mean = mean(damages)
pct_diff = round(100 * abs(mc_mean - ead_quadrature) / ead_quadrature; digits=1)
DataFrame(
    Method=["Stochastic Mean ($(n_samples) samples)", "EAD (Quadrature)"],
    Annual_Damage=[
        "\$$(round(mc_mean / 1e9; digits=3))B",
        "\$$(round(ead_quadrature / 1e9; digits=3))B"
    ]
)
```

```
]
)
```

Table 2: Comparison of stochastic mean damage vs. EAD from quadrature integration

Row	Method	Annual_Damage
	String	String
1	Stochastic Mean (500 samples)	\$10.249B
2	EAD (Quadrature)	\$9.478B

```
println("The stochastic mean differs from the EAD by about ${pct_diff}%.")
```

The stochastic mean differs from the EAD by about 8.1%.

Both methods estimate the same quantity (expected annual damage) — the stochastic mean via sampling and the EAD via numerical integration.

💡 Question 4

Why don't the stochastic mean and EAD match exactly? Which one would you trust more for a policy decision, and why?

8 Experiments: Change Things and See What Happens

Each experiment changes one parameter and shows the effect on risk. Read the code, run it, and answer the questions.

8.1 Experiment: Fat Tails

The GEV shape parameter ξ controls how heavy the right tail is. Let's see what happens when we increase it from 0.15 to 0.3.

Stochastic mode:

```
surge_dist_fat = GeneralizedExtremeValue(3.0, 1.0, 0.3)
surges_fat = rand(surge_dist_fat, n_samples)

damages_fat = map(surges_fat) do surge
    scenario = StochasticScenario(
        surges=[surge], mean_sea_level=[0.0], discount_rate=0.0
    )
    outcome = simulate(config, scenario, baseline)
    SimOptDecisions.value(outcome.damage)
end
mc_mean_fat = mean(damages_fat)
```

EAD mode:

```

ead_scenario_fat = EADScenario(
    surge_loc=3.0,
    surge_scale=1.0,
    surge_shape=0.3,
    mean_sea_level=[0.0],
    discount_rate=0.0
)
outcome_fat = simulate(ead_config, ead_scenario_fat, baseline)
ead_fat = SimOptDecisions.value(outcome_fat.expected_damage)

```

```

DataFrame(
    Scenario=["Baseline ( $\xi=0.15$ )", "Fat tails ( $\xi=0.3$ )"],
    Stochastic_Mean=[
        "\$$(round(mc_mean / 1e9; digits=3))B",
        "\$$(round(mc_mean_fat / 1e9; digits=3))B"
    ],
    EAD=[
        "\$$(round(ead_quadrature / 1e9; digits=3))B",
        "\$$(round(ead_fat / 1e9; digits=3))B"
    ]
)

```

Table 3: Effect of fatter tails ($\xi = 0.3$ vs. 0.15) on risk estimates

Row	Scenario	Stochastic_Mean	EAD
	String	String	String
1	Baseline ($\xi=0.15$)	\$10.249B	\$9.478B
2	Fat tails ($\xi=0.3$)	\$12.818B	\$14.549B

Question 5

How did increasing the shape parameter (fatter tails) affect the EAD? Why might the stochastic estimate be less reliable when the distribution has fatter tails?

8.2 Experiment: Sea-Level Rise

Now add 0.5 meters of sea-level rise to the baseline scenario.

EAD mode:

```

ead_scenario_slr = EADScenario(
    surge_loc=3.0,
    surge_scale=1.0,
    surge_shape=0.15,
    mean_sea_level=[0.5],
    discount_rate=0.0
)

```

```
outcome_slr = simulate(ead_config, ead_scenario_slr, baseline)
ead_slr = SimOptDecisions.value(outcome_slr.expected_damage)
```

```
pct_increase = round(100 * (ead_slr - ead_quadrature) / ead_quadrature; digits=1)
DataFrame(
  Scenario=["Baseline (MSL=0)", "SLR (MSL=+0.5m)"],
  EAD=[
    "\$(round(ead_quadrature / 1e9; digits=3))B",
    "\$(round(ead_slr / 1e9; digits=3))B"
  ],
  Change=["-", "+$(pct_increase)%"]
)
```

Table 4: Effect of 0.5m sea-level rise on Expected Annual Damage

Row	Scenario	EAD	Change
	String	String	String
1	Baseline (MSL=0)	\$9.478B	—
2	SLR (MSL=+0.5m)	\$12.743B	+34.4%

Question 6

How much did 0.5m of sea-level rise increase the EAD? Is the increase proportional to the SLR amount? What does this tell you about the relationship between sea-level rise and flood risk?

8.3 Experiment: Your City (from Wednesday)

Now it's your turn — use the city parameters you estimated on Wednesday. Replace the placeholder values for `H_city`, `D_city`, `W_city`, and `H_seawall` with your city's values, then run EAD to compare against Manhattan.

See the ICOW parameter documentation for the full list of parameters you can change — try adjusting others beyond the four listed here.

```
my_config = StochasticConfig(
  H_city=10.0,
  D_city=1500.0,
  W_city=30000.0,
  H_seawall=1.0
)
my_ead_config = EADConfig(
  n_years=1,
  integrator=QuadratureIntegrator(rtol=1e-6),
  H_city=my_config.H_city,
  D_city=my_config.D_city,
  W_city=my_config.W_city,
  H_seawall=my_config.H_seawall
)
```

```
)
outcome_my = simulate(my_ead_config, ead_scenario, baseline)
ead_my = SimOptDecisions.value(outcome_my.expected_damage)
```

```
DataFrame(
  City=["Manhattan (default)", "Your city"],
  EAD=[
    "\$(round(ead_quadrature / 1e9; digits=3))B",
    "\$(round(ead_my / 1e9; digits=3))B"
  ]
)
```

Table 5: Your city’s baseline risk compared to Manhattan

Row	City	EAD
	String	String
1	Manhattan (default)	\$9.478B
2	Your city	\$44.086B

💡 Question 7

How does your city’s baseline risk compare to Manhattan’s? What features of your city’s geometry (elevation, coastline length, seawall height) explain the difference?

Enrichment (Optional)

If you finish early, try one of these extensions:

- **VaR and CVaR:** Compute the 95th percentile damage (Value at Risk) and the average of damages exceeding that threshold (Conditional Value at Risk) from the stochastic samples
- **Damage function:** For each surge height from 0 to 15m (in 0.1m steps), compute the expected damage. Plot the resulting damage function
- **No defenses:** Try a true do-nothing policy ($a_{\text{frac}}=0.0$) — how does the scatter plot change? (Hint: what happens to the two branches?)
- **Stronger dike:** Try $a_{\text{frac}}=0.3$ — does doubling the dike height halve the EAD?

9 Summary

Key takeaways from this lab:

1. ICOW computes flood damage for a simplified coastal city using a wedge geometry
2. **Stochastic mode** shows individual outcomes — you see the full variability and tail risk
3. **EAD mode** integrates over the surge distribution — you get the expected value deterministically
4. Risk depends nonlinearly on surge height, GEV parameters, and city geometry

Your Answers

Write your answers in the spaces below. Each question references a specific section above — scroll up to review the context if needed.

Question 1 (Section 4: One Storm)

The histogram shows two clusters of damage values for a single surge height. Why does the same surge produce different damage amounts? What physical mechanism in the ICOW model creates this pattern?

 Your Answer

Replace this text with your answer.

Question 2 (Section 5: Many Storms)

Look at the scatter plot. You should see two “branches” — one where the dike holds and one where it fails. At what surge height do the branches merge? Also: what does the sparse sampling in the high-damage tail tell you about Monte Carlo efficiency?

 Your Answer

Replace this text with your answer.

Question 3 (Section 5: Risk Statistics)

A city planner says: “Our expected annual damage is \$X billion, so we should budget \$X billion per year for flood recovery.” What’s wrong with this reasoning?

 Your Answer

Replace this text with your answer.

Question 4 (Section 6: EAD Mode)

Why don’t the stochastic mean and EAD match exactly? Which one would you trust more for a policy decision, and why?

 Your Answer

Replace this text with your answer.

Question 5 (Section 7: Fat Tails)

How did increasing the shape parameter (fatter tails) affect the EAD? Why might the stochastic estimate be less reliable when the distribution has fatter tails?

💡 Your Answer

Replace this text with your answer.

Question 6 (Section 7: Sea-Level Rise)

How much did 0.5m of sea-level rise increase the EAD? Is the increase proportional to the SLR amount?

💡 Your Answer

Replace this text with your answer.

Question 7 (Section 7: Your City)

How does your city's baseline risk compare to Manhattan's? What features of your city's geometry explain the difference?

💡 Your Answer

Replace this text with your answer.

10 Submission

1. Write your answers in the Answer Template section above
2. **Render to PDF:**
 - In VS Code: Open the command palette (Cmd+Shift+P / Ctrl+Shift+P) → "Quarto: Render Document" → select Typst PDF
 - Or from the terminal: `quarto render index.qmd --to typst`
3. **Submit the PDF** to the Lab 4 assignment on Canvas
4. **Push your code** to GitHub (for backup):

```
git add -A && git commit -m "Lab 4 complete" && git push
```

Checklist

Before submitting:

- ☐ Packages load without error
- ☐ Single storm simulation runs and trace is inspected
- ☐ Distribution of damages for one surge height visualized (Figure 1)
- ☐ Stochastic simulation (many surges) produces surge vs. damage figure (Figure 2)
- ☐ Risk statistics computed (Table 1)
- ☐ EAD computed via quadrature and compared to stochastic mean (Table 2)

- ☐ Fat tails experiment completed (Table 3)
- ☐ Sea-level rise experiment completed (Table 4)
- ☐ Your city experiment completed (Table 5)
- ☐ All 7 questions answered in the Your Answers section
- ☐ Notebook renders to PDF without errors

Bibliography